

llmXive Automated Review of Qwen-Image-Agent: Bridging the Context Gap in Real-World Image Generation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the validity of citations supporting those claims.

1. Verification of “State-of-the-Art” Claims: The paper claims Qwen-Image-Agent achieves “state-of-the-art” performance on IA-Bench, MindBench, and WISE-Verified.

- **IA-Bench (Table 1):** The claim is supported. Qwen-Image-Agent (IA-score 45.4) outperforms the next best agentic model (SCOPE, 30.9) and closed-source models (Nano Banana Pro, 42.6). The margin is significant.
- **WISE-Verified (Table 2):** The claim is supported. Qwen-Image-Agent (0.9020) outperforms Nano Banana Pro (0.8760). However, the text should clarify that the closest competitor is a proprietary model, as the comparison between open-source agents and closed-source giants is a key nuance.
- **MindBench (Table 3):** The claim is **weakly supported**. The table shows Qwen-Image-Agent (0.42) vs. Nano Banana Pro (0.41). The difference is 0.01 (approx. 2.4%). While technically higher, claiming “state-of-the-art” based on such a marginal difference without statistical significance testing or error bars is an overstatement. The text also claims an “82.6% improvement” over Qwen-Image-2.0 (0.23), which is mathematically correct, but the leap to “SOTA” over the 0.41 baseline is not robustly justified by the data presented.

2. Citation and Model Existence:

- **Proprietary Models:** The paper cites `openai2025gptimage15` (GPT-Image-1.5) and `openai2024gptimage1` (GPT-Image-1). Given the paper’s date (2026), these model names and versions appear speculative or potentially hallucinated, as OpenAI’s public roadmap does not currently confirm a “GPT-Image-1.5” with the specific capabilities described. If these models do not exist or are not publicly available for evaluation, the comparative claims are factually unsupported. The authors must verify the existence of these specific model versions or replace them with confirmed baselines.
- **Benchmarks:** The citations for WISE (`Niu2025WISEAW`) and MindBench (`He2026MindBrushIA`) appear consistent with the provided bibliography, though the dates (2025/2026) suggest these are very recent or future-dated works. The authors should ensure these references correspond to actual, accessible preprints or publications.

3. Data Consistency:

- The ablation study (Table 4) claims that removing “Memory” drops the Memory score to 0.0. This is a strong claim that implies the agent has *no* memory capability without the specific

module. This is plausible but should be verified against the experimental setup description to ensure the “w/o Memory” condition didn’t inadvertently disable other context mechanisms that might contribute to memory-like behavior.

Recommendation: The paper requires minor revisions to temper the “state-of-the-art” claim on MindBench given the marginal performance difference and to verify the existence/naming of the proprietary baselines (GPT-Image-1.5). The core scientific claims regarding the “Context Gap” and the framework’s effectiveness on IA-Bench are well-supported by the data.

Action items.

- **[writing]** The claim of ‘state-of-the-art’ on MindBench is weak; Qwen-Image-Agent (0.42) only marginally beats Nano Banana Pro (0.41). Clarify if this difference is statistically significant or if SOTA refers to a specific subset.
- **[science]** Verify the existence of ‘GPT-Image-1.5’ (openai2025gptimage15). If this model does not exist or is not publicly available, the comparative claims are unsupported. Replace with confirmed baselines.
- **[writing]** The claim of SOTA on WISE-Verified (0.9020 vs 0.8760) is factually correct but lacks context. Explicitly acknowledge that the closest competitor is a proprietary model to avoid misinterpretation of open-source vs closed-source performance.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-organized collage of diverse generated images that effectively demonstrates the model’s capabilities as described in the caption. The visual quality is high, and the variety of subjects (characters, scenes, text) supports the claim of generating examples without visual references.

Figure 2

Figure 2 effectively demonstrates the model’s planning ability by displaying a clear, legible 4x4 grid of numbers that perfectly matches the specific spiral arrangement requested in the prompt. The visual output is uncluttered and directly supports the caption’s claim of solving the enumeration problem.

Figure 3

Figure 3 effectively demonstrates the model’s reasoning ability by showing the input maze, the reasoning process, and the correct output path. The visual components are clear, and the caption accurately describes the content.

Figure 4

The figure fails to support the caption’s claim of demonstrating web search ability, as it shows a static weather app mockup with a future date rather than a search or retrieval process.

Figure 5

Figure 5 effectively demonstrates the agent’s image search capability by visually mapping the user prompt to the specific ‘Gathered Context’ search results and the final generated image. The layout is clear, and the selected reference image is explicitly marked, making the workflow easy to follow.

Figure 6

Figure 6 effectively illustrates the feedback loop by displaying the initial user prompt, a failed generation with specific feedback on counting errors, and the corrected generation context and final image. The visual layout clearly demonstrates the agent’s ability to self-correct based on feedback, aligning perfectly with the caption’s claim.

Figure 7

Figure 7 effectively demonstrates the multi-image generation capability by displaying three distinct slides (Cover, Solar Power, Wind Power) that correspond to the detailed generation context provided in the diagram. The visual output aligns perfectly with the caption’s claim of splitting and allocating generation context, and the layout is clear and readable.

Figure 8

Figure 8 effectively illustrates the agent’s memory ability by displaying a conversation history, a memory selection step, and the final generated output. The visual layout clearly demonstrates how the model retrieves and utilizes specific context from previous turns to fulfill the current user request.

Figure 9

The figure provides a visual overview of the subtasks but fails to visually demonstrate the ‘4 tasks’ hierarchy mentioned in the caption, and the inner ring text labels are too small to read clearly.

Figure 10

Figure 10 presents a qualitative comparison of models across four tasks, but the ‘Grounded Context’ column for the Feedback task contains a significant contradiction between the visual output (5 red, 5 blue cars) and the accompanying feedback text (claiming 2 blue cars were missing).

Figure 11

Figure 11 provides a clear and comprehensive overview of the Qwen-Image-Agent framework, effectively visualizing the pipeline from user context to image generation. The diagram is well-structured with distinct sections for planning and grounding, and the internal labels are legible and consistent with the provided caption.

Action items.

- **[science]** Figure 4: The image displays a UI mockup of a weather generation task rather than a ‘web search’ process; it fails to visualize the retrieval of external knowledge or the search mechanism described in the caption.
- **[science]** Figure 4: The image contains a future date (August 14, 2025) and a generated 3D scene, which contradicts the claim of retrieving factual external knowledge from the web for a real-world query.
- **[science]** Figure 9: The central sunburst chart displays 17 segments (subtasks) but the caption claims ‘4 tasks’ without visually grouping these segments into the 4 parent categories, making the ‘4 tasks’ claim unverifiable from the figure alone.
- **[writing]** Figure 9: The text labels for the inner ring segments (e.g., ‘Math’, ‘Science’, ‘Game’) are extremely small and illegible, particularly in the top-left quadrant, hindering readability.
- **[science]** Figure 10: The ‘Grounded Context’ column displays a generated image of a parking lot with 5 red and 5 blue cars, which contradicts the ‘Feedback’ text box claiming the model failed to generate ‘2 blue cars’ (implying the correct count was 2). The visual evidence in the ‘Generated’ column does not match the textual evaluation in the ‘Feedback’ column.
- **[writing]** Figure 10: The ‘Grounded Context’ column header is split into ‘Generated’ and ‘Feedback’ sub-headers, but the ‘Generated’ sub-header sits above an image that appears to be the *correct* ground truth (5 red, 5 blue cars), while the ‘Feedback’ box describes a failure state. This labeling is confusing and likely inverted or misaligned with the data shown.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and shorthand that are not defined upon first use, creating barriers for non-specialist readers.

First, the acronym **MLLM** (Multimodal Large Language Model) is introduced in the “Implementation Details” section (sec/experiments.tex) and used repeatedly in the “Ablation Study” without ever being spelled out. Similarly, **DAG** (Directed Acyclic Graph) appears in the “High Latency and Cost” discussion (sec/experiments.tex) without definition.

Second, the term **IP** is used in the “Search-Driven Tasks” description (sec/benchmark.tex) to refer to “IP-related entities.” While common in tech circles, “Intellectual Property” or a more descriptive phrase like “famous characters” would be clearer for a general audience.

Third, the abbreviation **CoT** (Chain-of-Thought) is used in Table 1 (tables/ours.tex) for baseline names (e.g., “Bagel w/ CoT”) but is never defined in the table caption or the surrounding text.

Finally, there is a confusing inconsistency in the “Quantitative Results” section (sec/experiments.tex), where the authors state the framework improves the “Q-score,” despite the metric being explicitly defined and labeled as “IA-score” throughout the paper. This undefined “Q-score” term appears to be a typo or an undefined shorthand that needs correction.

Action items.

- **[writing]** Define ‘MLLM’ at first use. The acronym appears in ‘Implementation Details’ (sec/experiments.tex) and ‘Ablation Study’ without prior expansion, assuming reader familiarity with ‘Multimodal Large Language Model’.
- **[writing]** Replace ‘IP’ with ‘Intellectual Property’ or ‘famous characters’ in ‘Search-Driven Tasks’ (sec/benchmark.tex). ‘IP’ is industry jargon that may confuse non-specialist readers in a general benchmark description.
- **[writing]** Define ‘DAG’ in the ‘High Latency and Cost’ discussion (sec/experiments.tex). The term ‘DAG-based execution’ is used without expansion, assuming knowledge of ‘Directed Acyclic Graph’.
- **[writing]** Replace ‘CoT’ with ‘Chain-of-Thought’ in Table 1 (tables/ours.tex) and related text. The abbreviation is used in baseline names (e.g., ‘Bagel w/ CoT’) without definition in the table caption or main text.
- **[writing]** Clarify ‘Q-score’ in the ‘Quantitative Results’ section (sec/experiments.tex). The text states ‘improves the Q-score substantially’, but the metric is defined as ‘IA-score’ earlier. This inconsistency and undefined ‘Q-score’ term creates confusion.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent logical structure regarding the definition of the “Context Gap” and the proposed “Qwen-Image-Agent” framework. The premise that real-world prompts are underspecified (user context) and require external grounding (generation context) is well-established, and the proposed agentic loop logically follows as a mechanism to bridge this gap. The ablation studies generally support the claim that each component (Reason, Search, Memory, Feedback) contributes to the final performance, as removing them leads to performance degradation in their respective dimensions.

However, there are minor logical inconsistencies between the textual claims and the quantitative evidence presented in the tables. Specifically, the justification for the IA-score weighting scheme (Section 4.3) appears to downplay the importance of Memory (0.1 weight) despite the experimental results (Table 1) showing the proposed agent achieves its highest relative gains and absolute scores in the Memory dimension compared to baselines. This creates a slight disconnect between the authors’ stated prioritization of capabilities and the empirical evidence of where their framework shines.

Additionally, the discussion of the “Feedback” ablation (Section 5.3) characterizes the performance drop as “relatively smaller” and attributes it to the backbone’s strength. While the drop is indeed smaller than that of Search or Reason, the magnitude of the drop in Pass Rate (approx. 5-6 percentage points) is non-trivial. The text’s minimization of this drop without quantifying the relative percentage loss weakens the logical support for the claim that feedback is less critical in this specific configuration. Clarifying these points would strengthen the internal consistency of the argument.

Action items.

- **[science]** The IA-score formula assigns Memory a 0.1 weight, calling it ‘complementary,’ yet Table 1 shows the model achieves its highest relative gains in Memory. This contradicts the empirical finding that the framework excels here, suggesting the weighting scheme misaligns with the paper’s own conclusions about the framework’s strengths.
- **[writing]** In the Discussion, the text claims the ‘Feedback’ ablation drop is ‘relatively smaller’ due to strong rendering. However, Table 4 shows a ~12% relative drop in Pass Rate. The qualitative minimization of this drop is not supported by the magnitude of the quantitative data presented.
- **[writing]** The text claims an 82.6% improvement on MindBench over Qwen-Image-2.0. While the math (0.42 vs 0.23) is correct, the text refers to this as ‘performance’ without explicitly specifying it refers to the ‘Overall’ score in Table 3, which could cause ambiguity regarding sub-metric performance.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend beyond the immediate evidence provided in the text and tables, particularly regarding the magnitude of performance gains and the generalizability of the proposed framework.

First, the assertion of “state-of-the-art” performance on WISE-Verified (Section 5.2, Table 2) appears overstated. While Qwen-Image-Agent achieves the highest score (0.9020), the margin over the second-best model, Nano Banana Pro (0.8760), is approximately 2.6 percentage points. Given the nature of these benchmarks and the lack of reported standard deviations, confidence intervals, or statistical significance tests, declaring a definitive SOTA status is premature. The data suggests a competitive performance, but the leap to “state-of-the-art” implies a level of dominance not clearly evidenced by the raw numbers alone.

Second, the claim that the framework is “training-free” and broadly “compatible with existing image generators” (Introduction) risks overgeneralization. The ablation study in Table 4 reveals that the system’s performance is heavily dependent on the specific MLLM backbone used. When the default GPT-5.5-0424 is replaced with Qwen-Plus, the IA-score drops significantly (from 45.4 to 27.8). This suggests that the “agentic” success is not merely a result of the framework’s architecture but is inextricably linked to the superior reasoning capabilities of a specific, high-end proprietary model. The paper should temper its claims about the framework’s independence from the underlying model’s intelligence.

Finally, the motivation for IA-Bench relies on the premise that existing benchmarks “fail to systematically assess” agentic capabilities (Introduction). However, the paper utilizes MindBench and WISE for comparison, both of which are cited as evaluating reasoning and knowledge. The distinction between what IA-Bench measures (specifically “Plan” and “Memory” in a unified agentic loop) versus what MindBench measures needs to be more rigorously defined to avoid the impression that previous work was entirely blind to these dimensions. The current justification slightly overstates the gap in the literature.

Action items.

- **[science]** The claim that Qwen-Image-Agent achieves ‘state-of-the-art’ performance on WISE-Verified (Table 2) is overreaching. The reported score (0.9020) is only marginally higher than Nano Banana Pro (0.8760) and GPT-Image-1.5 (0.8250). Without statistical significance testing or error bars, asserting a definitive SOTA status over strong proprietary baselines is not fully supported by the data presented.
- **[science]** The paper claims the framework is ‘training-free’ (Introduction) and ‘compatible with existing image generators,’ yet the ablation study (Table 4) relies heavily on a specific, powerful MLLM backbone (GPT-5.5-0424). The performance drop when switching to Qwen-Plus suggests the ‘agentic’ success is tightly coupled with the specific capabilities of this proprietary model, potentially overgeneralizing the framework’s effectiveness to other, weaker backbones.
- **[writing]** The introduction states that existing benchmarks ‘fail to systematically assess’ agentic capabilities, justifying IA-Bench. However, the paper does not sufficiently address why

existing benchmarks like MindBench or WISE (which are used for comparison) are inadequate for the specific ‘Plan’ and ‘Memory’ dimensions, given that MindBench explicitly covers reasoning and knowledge. The distinction needs sharper justification to avoid overclaiming the novelty of the benchmark’s scope.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a novel agentic framework for image generation but raises several safety and ethical concerns that require clarification before acceptance.

First, the methodology relies heavily on external tools for “Context Grounding,” specifically the Google Search API and Jina API (Implementation Details, sec/experiments.tex). The manuscript does not address the privacy implications of sending user prompts to these third-party services, nor does it discuss how user data is handled, stored, or deleted. Furthermore, the retrieval of images from the web for “Grounding via Search” (sec/method.tex) introduces significant risks regarding copyright infringement and the potential generation of non-consensual deepfakes, particularly given the benchmark’s inclusion of “Celebrity” and “IP” tasks. The authors must explicitly state their compliance with search engine Terms of Service and outline any safeguards against generating harmful or infringing content.

Second, the construction of the IA-Bench benchmark involves human annotation and the use of LLMs to generate evaluation checklists (sec/benchmark.tex). The paper lacks details on the ethical treatment of human annotators, including whether they provided informed consent, were compensated fairly, and if the study received Institutional Review Board (IRB) approval. Given the involvement of human subjects in the data creation and evaluation pipeline, this information is critical for ethical compliance.

Finally, the system’s ability to retrieve and render images of real people (e.g., in the “Celebrity” subtask) poses a dual-use risk for creating misleading or harmful content. The authors should include a discussion on the potential for misuse and describe any technical or policy-level mitigations implemented to prevent the generation of non-consensual deepfakes or the violation of intellectual property rights.

Action items.

- **[writing]** The paper presents a novel agentic framework for image generation but raises several safety and ethical concerns that require clarification before acceptance. First, the methodology relies heavily on external tools for “Context Grounding,” specifically the Google Search API and Jina API (Implementation Details, sec/experiments.tex). The manuscript does not address the privacy implications of sending user prompts to these third-party services, nor does it discuss how user data is handled, stored,

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the claims of Qwen-Image-Agent is generally robust in terms of the breadth of benchmarks used (IA-Bench, WISE-Verified, MindBench) and the inclusion of ablation studies. However, several critical gaps in experimental rigor and statistical reporting prevent a full acceptance of the results as definitive.

First, the evaluation methodology relies exclusively on VLM-based automated checklists (Section 4.2, “Evaluation Criterion”). While the authors mention human annotation for benchmark construction, there is no reported inter-rater reliability or validation of the VLM evaluator against human ground truth. In complex agentic tasks involving reasoning and memory, VLMs can exhibit systematic biases or hallucinations in their own evaluation. Without a human validation study (e.g., reporting Cohen’s kappa between VLM and human scores on a subset of 50-100 samples), the absolute Pass Rate numbers (e.g., 45.4% IA-score) lack a verified scale of accuracy.

Second, the ablation study results in Table 4 (sec/tables/ablate.tex) present a statistical anomaly. The “w/o Memory” condition reports a 0.0% Pass Rate for the Memory dimension. In a well-functioning system, removing a module should degrade performance, not cause a total collapse to zero unless the system fails to execute the task entirely (e.g., a crash or infinite loop). The authors must clarify whether this is a hard failure mode of the agent or a metric artifact. If the agent simply cannot complete the task without memory, the metric should reflect a “task completion” failure rather than a “checklist accuracy” of zero, or the experimental setup needs to be adjusted to ensure the agent attempts the task even with degraded context.

Third, the comparison with proprietary baselines (Table 1, sec/tables/ours.tex) is potentially confounded. The paper states that “all agentic generation baselines are evaluated under the same experimental setting,” but it is unclear if the proprietary models (GPT-Image-1.5, Nano Banana) were forced to run through the Qwen-Image-Agent pipeline (using GPT-5.5 as the planner) or if they were evaluated in their native, direct generation mode. If the proprietary models were run directly while Qwen-Image-Agent used its full agentic pipeline, the performance gap may be attributed to the *agentic framework* rather than the *image generation model* itself. A fair comparison requires either running all models through the same agentic pipeline or clearly distinguishing between “Direct” and “Agentic” modes for every baseline.

Finally, the sample size for the benchmarks is stated as 730 instances (Section 3.1). While this is reasonable, the paper does not report confidence intervals or standard deviations for the reported metrics. Given the stochastic nature of both the image generation and the VLM evaluation, providing confidence intervals (e.g., via bootstrapping) would strengthen the claim that the observed improvements are statistically significant and not due to random variance.

Action items.

- **[science]** The ablation study in Table 4 (sec/tables/ablate.tex) reports a 0.0% Pass Rate for the ‘w/o Memory’ condition on the Memory dimension. This absolute failure suggests a potential implementation error (e.g., the agent crashing or skipping the task entirely) rather than a graceful degradation of capability. The authors must clarify if this is a hard failure mode or a metric calculation artifact, as it undermines the validity of the comparison.
- **[science]** The evaluation protocol relies entirely on VLM-based checklists (sec/benchmark.tex, ‘Evaluation Criterion’). The paper lacks a human-in-the-loop validation study to estimate the agreement rate (Cohen’s kappa) between the VLM evaluator and human annotators. Without this, the reported Pass Rates and IA-scores may reflect VLM biases rather than true generation quality, especially for complex reasoning tasks.
- **[science]** The baseline comparison in Table 1 (sec/tables/ours.tex) mixes proprietary API models (e.g., GPT-Image-1.5) with open-source models. The paper does not specify if the proprietary baselines were evaluated with the same ‘agentic’ pipeline (using the same MLLM backbone and search tools) or if they were run in their native ‘direct’ mode. If the latter, the comparison is confounded by the agent framework itself rather than the model’s inherent capabilities.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis presented in the paper is generally sound in its structure but lacks necessary rigor regarding uncertainty quantification and metric definition consistency.

First, the definition of the **IA-score** in Section 4.3 (“Evaluation Criterion”) is ambiguous regarding the input metric. The formula is defined as a weighted sum of “micro average evaluation scores,” yet Table 1 (tables/ours.tex) reports both **Pass Rate (PR)** and **Checklist Accuracy (CA)**. It is unclear whether the reported IA-score of 45.4 is derived from the PR or CA columns. Given that PR is a stricter metric (requiring all checklist items to pass) and CA is an average, the choice significantly impacts the aggregate score. The authors must explicitly state which metric feeds into the IA-score calculation to ensure the results are reproducible.

Second, the **ablation studies** (Table 4, tables/ablate.tex) present point estimates for performance drops (e.g., Memory dimension dropping to 0.0% without memory context) but provide no measure of statistical significance. In benchmarks involving VLM-based evaluation, there is inherent stochasticity in the evaluation model’s judgments. Without reporting confidence intervals (e.g., 95% CI via bootstrapping) or p-values from a significance test (e.g., McNemar’s test for paired binary outcomes), it is difficult to assert that the observed differences are robust and not artifacts of the specific evaluation run. The claim that “removing any grounded context leads to a clear drop” is qualitatively true but statistically unverified.

Finally, the **evaluation protocol** relies entirely on a single VLM (GPT-5.5-0424) to determine the binary pass/fail status of 1801 checklist items. The paper does not discuss the potential bias or variance introduced by this single-evaluator approach. Standard practice in such benchmarks often involves reporting inter-rater reliability (e.g., Cohen’s Kappa) if multiple evaluators are used, or at least acknowledging the limitation of a single-model evaluator. The absence of any error bars or uncertainty estimates on the reported percentages (e.g., 45.3% Pass Rate) limits the statistical interpretability of the “state-of-the-art” claims.

Action items.

- **[writing]** The statistical analysis presented in the paper is generally sound in its structure but lacks necessary rigor regarding uncertainty quantification and metric definition consistency. First, the definition of the IA-score in Section 4.3 (“Evaluation Criterion”) is ambiguous regarding the input metric. The formula is defined as a weighted sum of “micro average evaluation scores,” yet Table 1 (tables/ours.tex) reports both Pass Rate (PR) and Checklist Accuracy (CA). It is unclear whether the reporte

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high level of clarity and professional flow, effectively communicating the proposed “Context Gap” framework and the Qwen-Image-Agent architecture. The narrative structure is logical, moving smoothly from problem identification to methodological formulation, benchmark creation, and experimental validation. The use of bolding for key terms (e.g., **Context Gap**, **Context-Aware Planning**) aids readability and helps the reader track the core concepts throughout the text.

However, there are several minor but distinct grammatical errors and typos that detract from the polish of the final draft. In the Introduction, specifically within the contributions list, the fourth bullet point contains a subject-verb agreement error: “Qwen-Image-Agent... achieve state-of-the-art performance” should read “achieves.” Similarly, in Section 3.3, the phrase “routes each questions” incorrectly pairs a singular determiner with a plural noun; this should be corrected to “routes each question.”

In the Results section (Section 5.1), the word “highlights” is misspelled and should be “highlights.” Furthermore, in the Discussion section (Section 5.4), the sentence “we adopt a stronger MLLM backbone and substantially improves” suffers from a subject-verb disagreement; the verb should be “improve” to agree with the subject “we.”

Finally, a technical writing issue exists in the Appendix where two distinct figures (the multi-image case and the feedback case) are assigned the exact same LaTeX label (`\label{fig:case_feedback}`). This will result in broken or incorrect cross-references in the compiled document. The authors should ensure unique labels for all figures and tables. Addressing

these specific points will significantly improve the manuscript’s overall quality.

Action items.

- **[writing]** In the Introduction (contributions list, item 4), correct the subject-verb agreement error: ‘Qwen-Image-Agent... achieve’ should be ‘achieves’ to match the singular subject.
- **[writing]** In Section 3.3 (Context-Aware Planning), the sentence ‘routes each questions’ contains a number agreement error. It should be ‘routes each question’ or ‘routes the questions’.
- **[writing]** In Section 5.1 (Quantitative Results), the word ‘hightlights’ is a typo and should be corrected to ‘highlights’.
- **[writing]** In Section 5.4 (Discussion, Unidentified Context Gaps), the final sentence ‘we adopt a stronger MLLM backbone and substantially improves’ has a subject-verb agreement error. It should be ‘improve’ to match the plural subject ‘we’.
- **[writing]** In the Appendix, Figure 6 (multi-image case) and Figure 5 (feedback case) share the identical label ‘\label{fig:case__feedback}’. This will cause reference conflicts in the compiled PDF. One label must be renamed (e.g., to ‘fig:case__multiimage’).